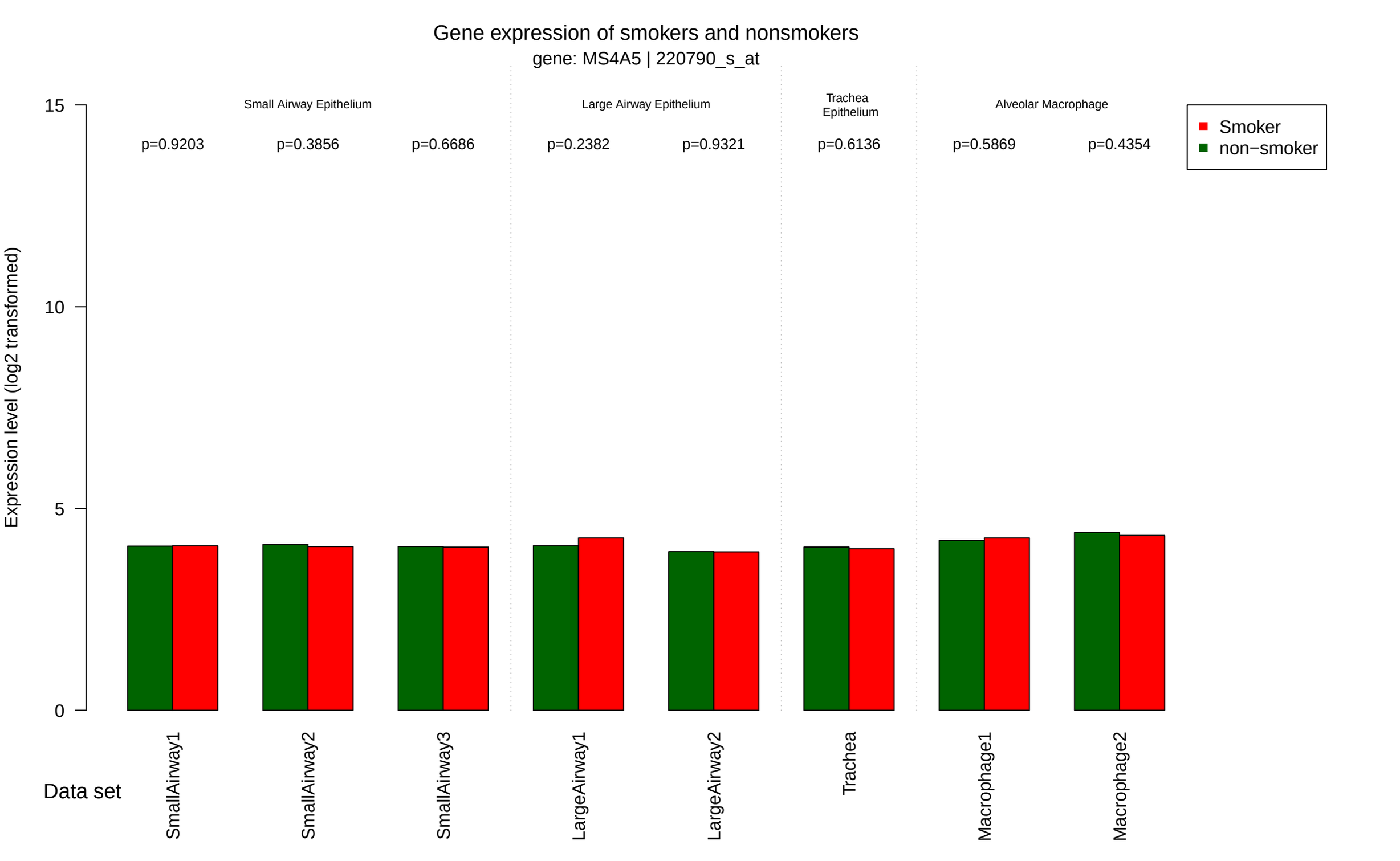
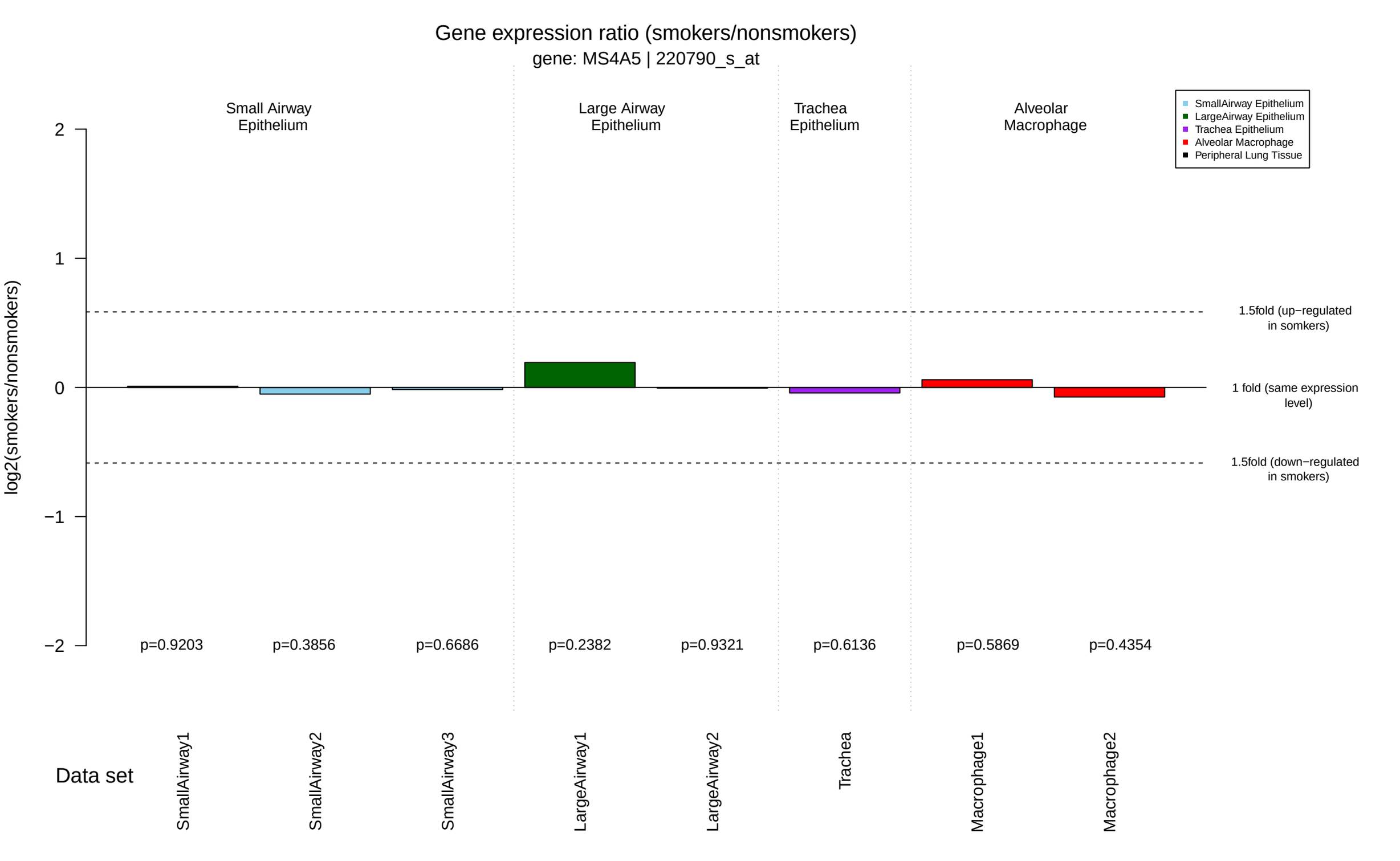
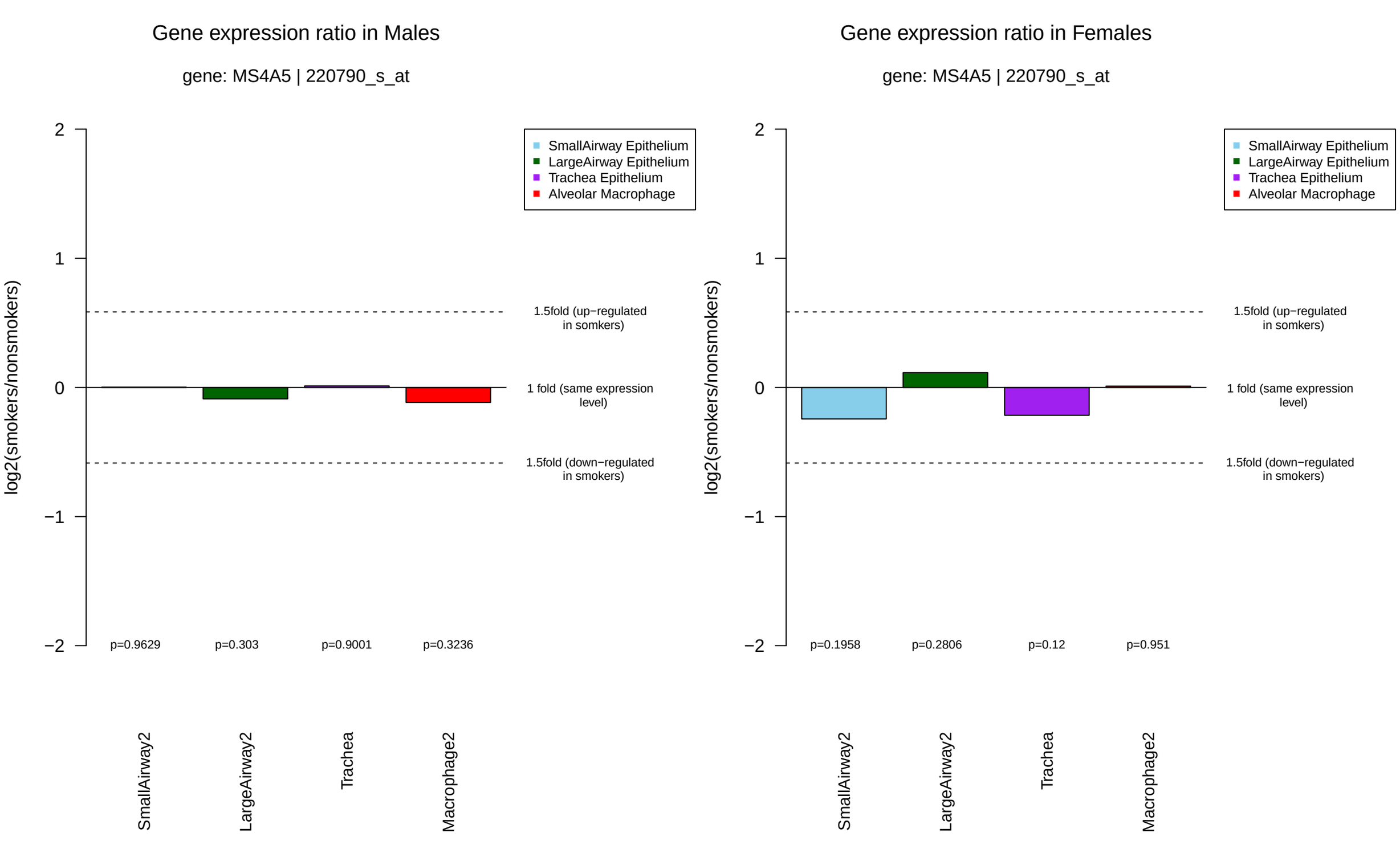
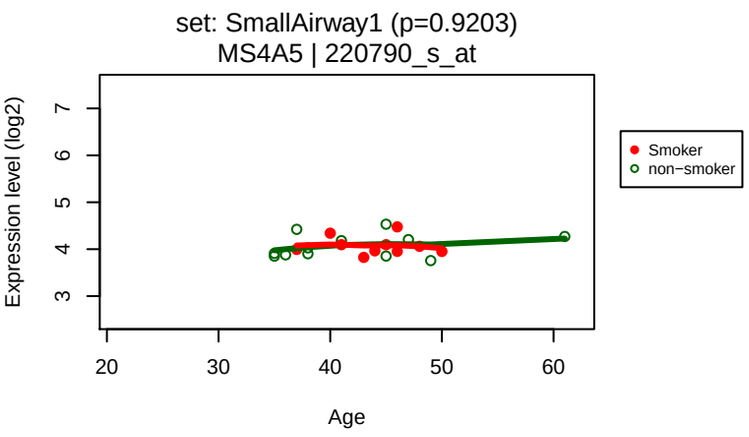
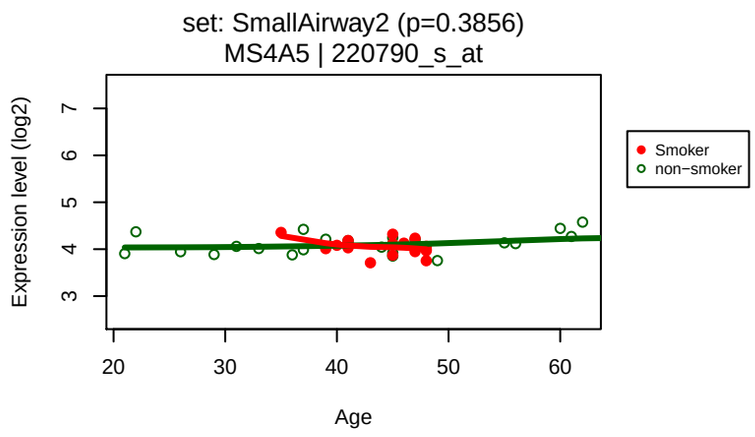
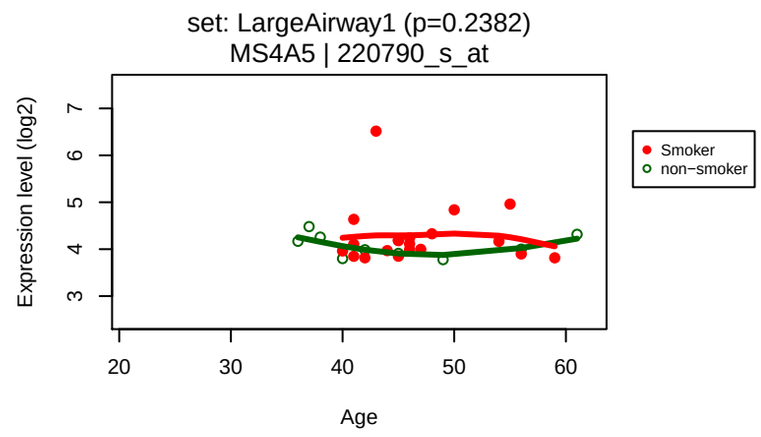

Gene expression of smokers and nonsmokers
gene: MS4A5 | 220790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9203
p=0.3856
p=0.6686
p=0.2382
p=0.9321
p=0.6136
p=0.5869
p=0.4354
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

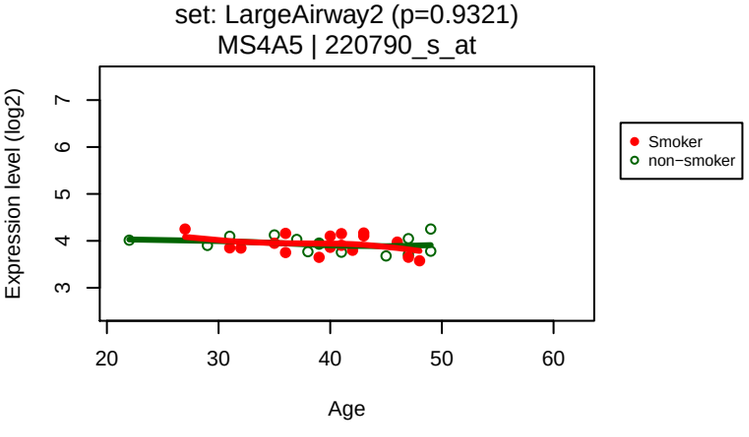

Gene expression ratio (smokers/nonsmokers)
gene: MS4A5 | 220790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9203
p=0.3856
p=0.6686
p=0.2382
p=0.9321
p=0.6136
p=0.5869
p=0.4354
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

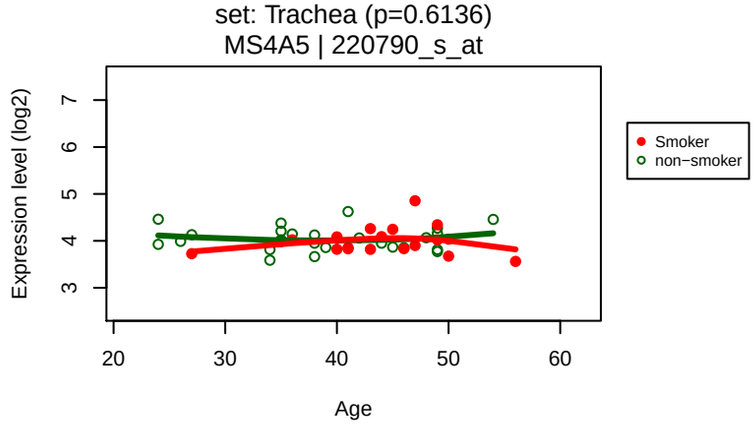

Gene expression ratio in Males
gene: MS4A5 | 220790_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9629
p=0.303
p=0.9001
p=0.3236
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A5 | 220790_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1958
p=0.2806
p=0.12
p=0.951
SmallAirway2
LargeAirway2
Trachea
Macrophage2

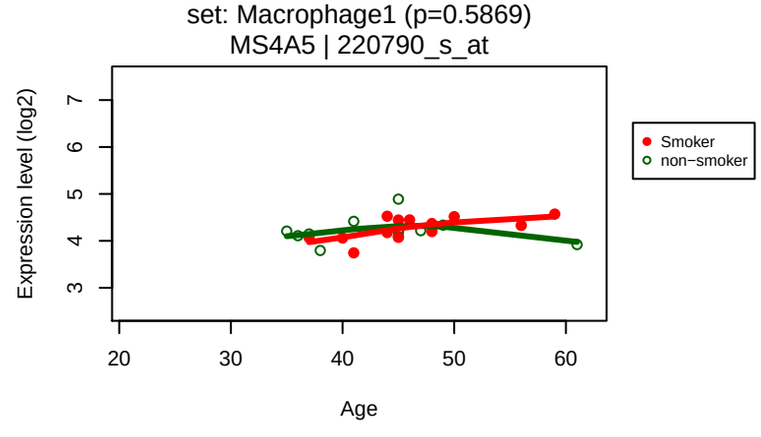

set: SmallAirway1 (p=0.9203)
MS4A5 | 220790_s_at
Expression level (log2)
Age
Smoker
non−smoker

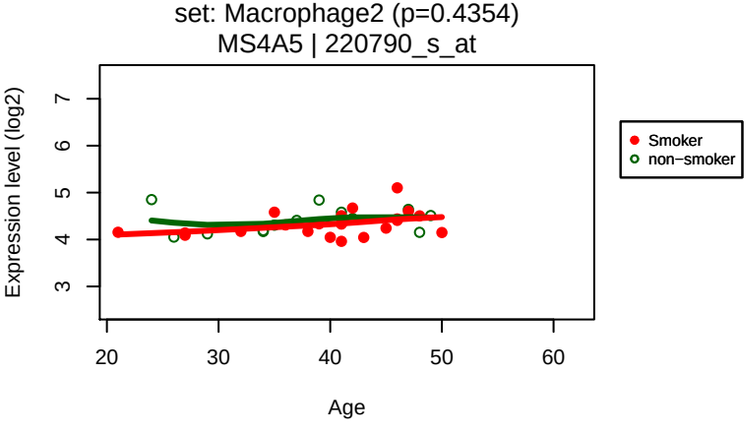

set: SmallAirway2 (p=0.3856)
MS4A5 | 220790_s_at
Expression level (log2)
Age
Smoker
non−smoker

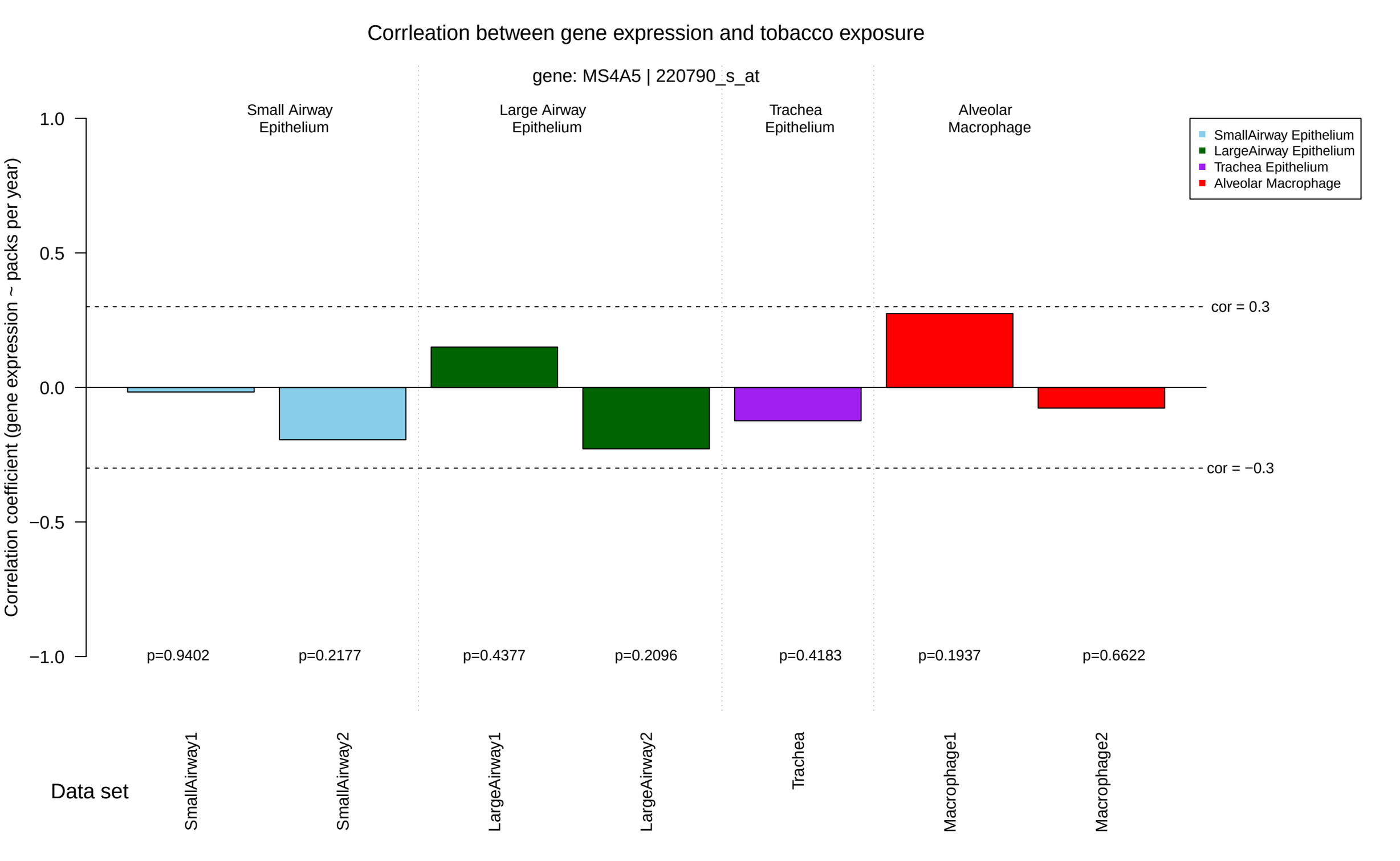

set: LargeAirway1 (p=0.2382)
MS4A5 | 220790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9321)
MS4A5 | 220790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6136)
MS4A5 | 220790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5869)
MS4A5 | 220790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4354)
MS4A5 | 220790_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A5 | 220790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9402
p=0.2177
p=0.4377
p=0.2096
p=0.4183
p=0.1937
p=0.6622
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2